

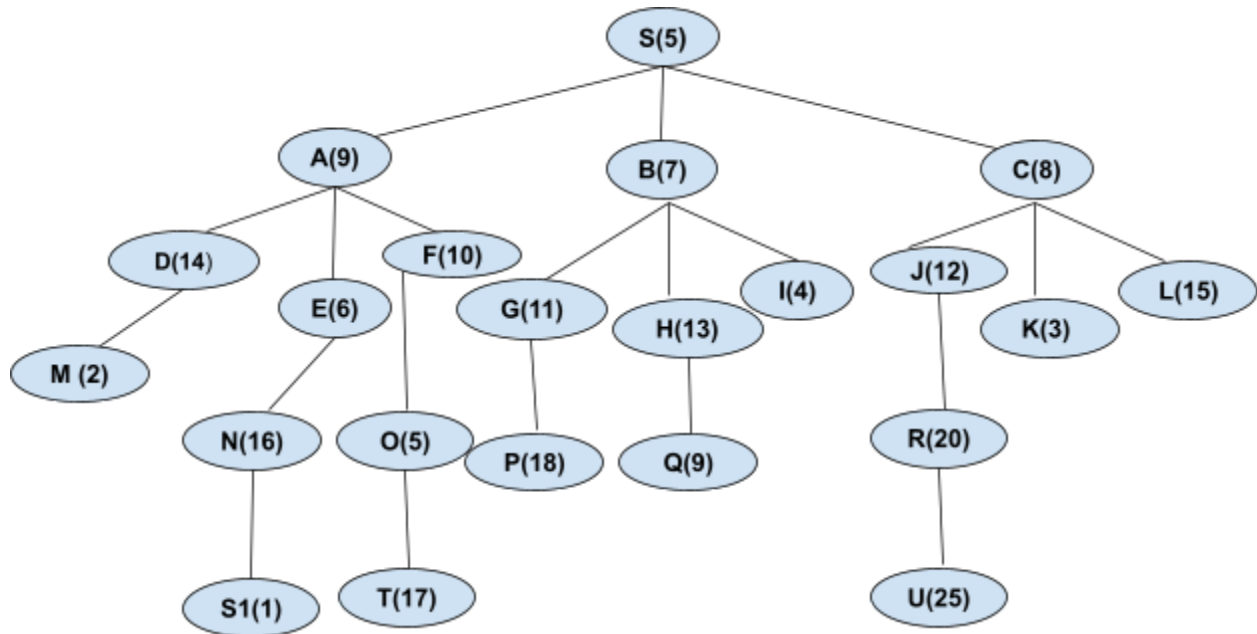


Instructor : Bushra Mehmood

Quiz 03

Given State Space (Maximize $h(n)$)

PART A — Local Beam Search



Use:

1. Beam width $k = 2$
2. Beam width $k = 3$

Tasks:

- Run Local Beam Search step-by-step
- Show successor pool at each iteration
- Identify discarded branches
- Does it reach U (25)? Why or why not

PART B — Simulated Annealing

Now apply Simulated Annealing with:

- Initial state = A (9)
- Temperature schedule:
- $T_0 = 10$
- Cooling: $T = T - 2$ each step

Why can simulated annealing reach $U(25)$?